$\equiv \exists \varepsilon > 0 \ \forall \delta > 0 \ \exists x ((0 < |x-a| < \delta) \wedge (|f(x) - L| \geq \varepsilon))$

Because the statement "$\lim_{x \to a} f(x)$ does not exist" means for all real numbers L, $\lim_{x \to a} f(x) \neq L$, this can be expressed as

$$\boxed{\forall L \exists \varepsilon > 0 \ \forall \delta > 0 \ \exists x (0 < |x-a| < \delta \wedge |f(x) - L| \geq \varepsilon)}.$$

This last statement says that for every real number L there is a real number $\varepsilon > 0$ such that for every real number $\delta > 0$, there exists a real number x such that $0 < |x-a| < \delta$ and $|f(x) - L| \geq \varepsilon$.

Exercises 1.5

1. Translate these statements into English, where the domain for each variable consists of all real numbers.

a) $\forall x \exists y (x < y)$: For all real numbers x, there exists a real number y which is larger than x. Simplified: For any arbitrary real number, there exists a larger real number.

b) $\forall x \forall y (((x \geq 0) \wedge (y \geq 0)) \longrightarrow (xy \geq 0))$

For all real numbers x and for all real numbers y, if both x and y are ~~positive~~ non-negative, then their product xy is non-negative.

Simplified: The product of any two arbitrary non-negative real numbers is non-negative.

c) $\forall x \forall y \exists z (xy = z)$

For all real numbers x and for all real numbers y, there exists a real number z which is equal to the product xy.

Simplified: The product of any two arbitrary real numbers exists and is a real number.

2. Translate these statements into English, where the domain for each variable consists of all real numbers.

a) $\exists x \forall y (xy = y)$: There exists a real number x such that for all real numbers y, the product xy is equal to y.